

capture on trading volume priced greater than \$1.00.¹⁸¹⁵ This alternative may also result in complications for orders priced below \$1. Specifically, for orders priced below \$1 this alternative would lead to one of two outcomes that could have negative effects for stocks trading right at the \$1.00 threshold. For these stocks, the minimum pricing increment is \$0.0001 regardless of the access fee applied. If the fee cap for sub-\$1 orders were to be kept proportional at 0.30% and 0.15% for trades priced above \$1.00, then if those stocks prices drop below \$1.00 this would result in a situation where there was a group of stocks with the same tick size (i.e., \$0.0001) but two different fee caps. This could place stocks with the higher fee cap at a competitive disadvantage relative to the stocks with the lower access fee cap. This outcome is also more complicated than both the baseline and adopted rule which both have at most one fee cap per tick size. Alternatively, if the fee cap for orders priced below \$1 were to be set uniformly at 0.30%, then this alternative would create a discontinuity in the cost of accessing liquidity at the \$1.00 price threshold for stocks assigned the 15 mils fee cap; likewise, if the fee cap for orders priced below \$1 were to be set uniformly at 0.15%, then it would create a discontinuity at the \$1.00 price point for stocks with the 30 mil fee cap. This discontinuity could create distortions in liquidity provision as discussed in section VII.D.2.c.¹⁸¹⁶ Consequently, this alternative results in either a situation in which there are two fee caps for the \$0.0001 tick, or there exists a discontinuity in the cost of accessing liquidity at the \$1 price point causing distortions in liquidity provision. Thus, this alternative appears to create costs without corresponding benefits.

¹⁸¹⁵ See supra section VII.D.2.b

¹⁸¹⁶ Specifically, see supra note 669, on the issue of discontinuities in the cost of accessing liquidity near the \$1 threshold.

One commenter suggested the Commission plan to further study the question of access fee caps in combination with the change in the tick.¹⁸¹⁷ For the reasons discussed above and elsewhere in this release, the Commission adopts the 10 mil access fee cap. Commission staff, however, will review and study the effects of the amendments as described in section VII.D.

b. Higher or Lower Uniform Access Fee Cap

The Commission could have adopted different uniform access fee caps. An access fee cap must stay below 50% of the minimum pricing increment in order to preserve price coherence.¹⁸¹⁸ Alternatively, if the access fee cap is set below an exchanges' net capture rate, then it can adversely affect existing exchange pricing practices.¹⁸¹⁹ Some commenters suggested retaining the current uniform 30 mils cap for stocks with prices above \$1.00.¹⁸²⁰ A uniform 30 mils level would be above 50% of the minimum pricing increment under the adopted amendments, and thus would not preserve price coherence. Another commenter suggested a uniform access fee cap below 10 mils.¹⁸²¹

As discussed in section VII.D.2.b, exchanges have sufficient flexibility under the adopted amendments to maintain their current net-capture and agency business models on stocks with prices above \$1. A uniform access fee higher than 10 mils would afford exchanges more

¹⁸¹⁷ Nasdaq Letter IV at 11. This commenter recommended this alternative as phase 1 in a three-phase data gathering process where, in phase 2, the Commission would collect a year of data from phase 1's changes and then consider a further access fee cap reduction for stocks with a \$0.005 tick. In phase 3, the Commission would collect an additional year of data to consider a lower fee cap for stocks with a \$0.01 tick. *Id.* at 2 and 11.

¹⁸¹⁸ See supra section VII.D.2.a.

¹⁸¹⁹ If the access fee cap was set below an exchange's net capture rate, then its profitability would decrease because it would no longer be able to charge fees high enough to cover any non-negative rebate. To retain the same net capture rate, the exchange would have to charge a negative rebate (i.e., a fee), which would represent a major change in pricing model.

¹⁸²⁰ See Citigroup Letter at 6, WFE Letter at 4.

¹⁸²¹ See IEX Letter IV at nn.14 and 21.

flexibility relative to the adopted amendments. For stocks with prices less than \$1.00, an access fee percentage (of the share price) higher than the adopted 0.1% would imply, relative to the adopted amendments, a lower revenue loss on sub \$1.00 trading.¹⁸²² To illustrate with an example, if the access fee percentage were 0.15% instead of the adopted 0.10%, then exchanges' expected revenue loss on sub \$1.00 trading would be approximately \$41 million across those exchanges charging the full 0.30% under the baseline, instead of approximately \$55 million under the adopted amendments.¹⁸²³ The main cost of an access fee cap above 10 mils would be to increase transaction costs for stocks with economic spreads smaller than the minimum pricing increment.¹⁸²⁴ Also, an access fee cap higher than 10 mils would allow for a greater wedge to exist between displayed prices and the net prices that are actually realized, potentially undermining price transparency.

In contrast, if the access fee caps were set below the adopted levels, then the effects described in the prior paragraph would all flip. Namely, relative to the adopted amendments, transaction costs for stocks with economic spreads smaller than the minimum pricing increment would be lower, and there could exist a smaller wedge between displayed prices and the net prices that are actually realized, potentially improving price transparency. However, relative to the adopted amendments, exchanges could no longer have sufficient flexibility to earn their net capture on stocks with prices above \$1.00, and exchanges would incur a greater loss in revenue on sub \$1.00 trading.

¹⁸²² See supra section VII.D.2.b for a discussion of lost revenue under the adopted amendments.

¹⁸²³ See Table 14 in supra section VII.D.2.b. In that table, if the access fee percentage were 0.15% instead of the adopted 0.10% then the lost revenue on sub \$1.00 trading would be approximately \$41 million across exchanges charging the full 0.30% under the baseline.

¹⁸²⁴ For stocks with wider economic spreads, the higher access fee would most likely reduce the spread in equilibrium, implying little or no effect on transaction costs. See supra sections VII.B.3 and VII.D.2.c

An access fee cap higher than the adopted 10 mils, and the associated higher rebates, also exacerbates the potential conflict of interest for broker-dealers who route customers' orders to the exchanges. As discussed in section VII.D.3, fees and rebates introduce the potential for a conflict of interest if those fees and rebates are not fully passed through to the routing broker-dealers' customers. A higher access fee cap would increase the potential proceeds a broker-dealer would receive if it acted on the conflict of interest. A lower access fee cap would decrease the differences between the fees and rebates offered by different exchanges, which would decrease the potential proceeds a broker-dealer would receive if it acted on the conflict of interest.

Additionally, relative to the adopted amendments to Rule 610(c), a lower access fee cap could hinder an exchange's ability to differentiate itself from other exchanges on the basis of its pricing schedule, whereas a higher access fee cap could enable more differentiation. As discussed in section VII.E.2.b, the Commission expects that exchanges will continue to set fees and rebates at or near the access fee cap; therefore, a higher or lower access fee cap would likely have minimal effect on pricing differentiation across exchanges. For retail investors, an access fee cap different from the adopted levels would likely have little effect on retail market quality for reasons discussed earlier.¹⁸²⁵ With regard to exchange trading versus off-exchange trading, as discussed above,¹⁸²⁶ an access fee cap lower than the adopted level could bring more trading volume onto exchanges by further relieving tick constraints that drive volume off-exchange. A higher access fee cap would reverse these effects.

¹⁸²⁵ See supra note 1480 and surrounding text.

¹⁸²⁶ See supra section VII.E.2.b.

VIII. Paperwork Reduction Act

Certain provisions of the rules and rule amendments contain “collection of information requirements” within the meaning of the Paperwork Reduction Act of 1995 (“PRA”).¹⁸²⁷ The Commission requested comment on the collection of information requirements in the Regulation NMS Proposal and submitted relevant information to the Office of Management and Budget (“OMB”) for review in accordance with 44 U.S.C. 3507(d) and 5 CFR 1320.11. The title of the new collection of information is “Odd-Lot Information Acceleration.” An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the agency displays a currently valid control number. The Commission has received an OMB control number (3235-0802) for this collection of information.

One commenter stated that the hourly rates for certain positions were inconsistent across the four proposals related to separate aspects of equity market structure and Regulation NMS.¹⁸²⁸ No other comments were received discussing the PRA. The hourly rates used to monetize burden hours differ across releases in order to account for changes in inflation rates. Consistent with this approach, the hourly rate figures discussed below have been updated from those cited in the Proposing Release to reflect recent inflation rates. In addition, certain estimates outlined in the MDI Adopting Release have been modified, as discussed in section VII.G below, to conform to the adopted amendments.

¹⁸²⁷ 44 U.S.C. 3501 et seq.

¹⁸²⁸ See Data Boiler Letter I at 16 (identifying what it termed “inconsistent rates” for the Attorney and Compliance Manager positions).

A. Summary of Collection of Information

The rule amendments include a collection of information within the meaning of the PRA. Specifically, the amendments to Rule 603(b) require the exclusive SIPs to collect, consolidate, and disseminate odd-lot information, including the best odd-lot orders to buy and sell. The exclusive SIPs are also required to disseminate indicators of the applicable round lot size and minimum pricing increment for each NMS stock, both of which will be provided to the exclusive SIPs by the primary listing exchange.

B. Proposed Use of Information

The information collected under the amendments to Rule 603(b) will be consolidated and disseminated by the exclusive SIPs to market participants who will use this odd-lot information for trading. Widespread availability of odd-lot information promotes fair and efficient markets and facilitates the ability of brokers and dealers to trade more effectively and to provide best execution to their customers. The round lot and minimum pricing increment indicators that will be disseminated by the exclusive SIPs will provide market participants with information about the parameters for trading in a particular NMS stock.

C. Respondents

The collection of information under amended Rule 603(b) will apply to the two exclusive SIPs.

D. Total Annual Reporting and Recordkeeping Burden

1. Initial Burden Hours and Costs

The two exclusive SIPs will have to modify their systems to collect, consolidate, and disseminate the odd-lot information, including the best odd-lot orders to buy and sell, that they

do not currently collect, consolidate, and disseminate¹⁸²⁹ and to disseminate the round-lot and minimum pricing increment indicators provided by the primary listing exchange. These modifications will involve the addition of new hardware, network infrastructure, and bandwidth, as well as programming and development costs, to take in additional inbound odd-lot quotation messages from SROs, to calculate odd-lot information, and to consolidate and disseminate odd-lot information and the round lot and minimum pricing increment indicators to subscribers.

The Commission estimates that each exclusive SIP will incur 440 initial burden hours to modify its systems to collect, calculate, consolidate and disseminate odd-lot information and to disseminate the round-lot and minimum pricing increment indicators¹⁸³⁰ and initial external costs of \$412,500 to purchase the necessary technology to effect such modifications.¹⁸³¹ Thus, the

¹⁸²⁹ The exclusive SIPs currently disseminate odd-lot transaction data.

¹⁸³⁰ The Commission estimates the monetized initial burden for this requirement to be \$167,670, broken down as follows: [(Sr. Programmer at \$399/hour for 210 hours) + (Sr. Systems Analyst at \$343/hour for 180 hours) + (Compliance Manager at \$373/hour for 20 hours) + (Director of Compliance at \$588/hour for 10 hours) + (Compliance Attorney at \$440/hour for 20 hours)] = 440 initial burden hours to modify its systems to comply with the requirement to collect, calculate, and disseminate odd-lot information. The Commission based these estimates on 10% of the initial burden hour estimates for each exclusive SIP to become a competing consolidator provided in the MDI Rules to account for the fact that these amendments do not require the exclusive SIPs to calculate and disseminate full consolidated market data (e.g., depth of book data or auction information) as defined in the MDI Rules. See MDI Adopting Release, supra note 10, at 18712-13. The Commission derived the hourly rate figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

¹⁸³¹ The Commission arrived at this estimate by dividing the initial external cost estimate provided in the MDI Rules for each exclusive SIP to become a competing consolidator by three to account for the fact that the exclusive SIPs would not need to build aggregation systems in three separate data centers to collect, calculate, and disseminate odd-lot information. See MDI Adopting Release, supra note 10, at 18712-13.

Commission estimates that the total initial burden hours for two exclusive SIPs will be 880 burden hours¹⁸³² and that total initial external costs would be \$825,000.¹⁸³³

2. Ongoing Burden Hours and Costs

The Commission believes that the two exclusive SIPs will incur annual ongoing burden hours and external costs to operate and maintain their modified systems to collect, calculate, and disseminate odd-lot information and to disseminate the round-lot and minimum pricing increment indicators. The Commission estimates that each exclusive SIP will incur 132 ongoing, annual burden hours¹⁸³⁴ and ongoing, annual external costs of \$123,725 to operate and maintain its systems to collect, calculate, and disseminate odd-lot information and to disseminate the round-lot and minimum pricing increment indicators.¹⁸³⁵ Thus, the Commission estimates that

¹⁸³² The Commission estimates the monetized initial burden for this requirement to be \$335,340, broken down as follows: [(Sr. Programmer at \$399/hour for 210 hours) + (Sr. Systems Analyst at \$343/hour for 180 hours) + (Compliance Manager at \$373/hour for 20 hours) + (Director of Compliance at \$588/hour for 10 hours) + (Compliance Attorney at \$440/hour for 20 hours)] x [(2 exclusive SIPs)] = 880 total initial burden hours across the exclusive SIPs.

¹⁸³³ The Commission estimates total initial external costs as follows: initial external costs of \$412,500 per exclusive SIP x (2 exclusive SIPs) = \$825,000.

¹⁸³⁴ The Commission estimates the monetized annual ongoing burden for this requirement to be \$50,301, broken down as follows: [(Sr. Programmer at \$399/hour for 63 hours) + (Sr. Systems Analyst at \$343/hour for 54 hours) + (Compliance Manager at \$373/hour for 6 hours) + (Director of Compliance at \$588/hour for 3 hours) + (Compliance Attorney at \$440/hour for 6 hours)] = 132 ongoing, annual burden hours to operate and maintain its systems to comply with the requirement to collect, calculate, and disseminate odd-lot information. The Commission based these estimates on 10% of the ongoing, annual burden hour estimates provided in the MDI Rules for each exclusive SIP competing consolidator to operate and maintain its systems to comply with Rules 614(d)(1) through (4) to account for the fact that these amendments do not require the exclusive SIPs to calculate and disseminate full consolidated market data (e.g., depth of book data or auction information) as defined in the MDI Rules. See MDI Adopting Release, supra note 10, at 18712-13. The Commission derived the hourly rate figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

¹⁸³⁵ The Commission arrived at this estimate by dividing by three the ongoing, annual external cost estimate provided in the MDI Rules for each exclusive SIP competing consolidator to operate and maintain its systems to comply with rules 614(d)(1) through (4) to account for the fact that the exclusive SIPs will not need to build aggregation systems in three separate data centers to collect, calculate, and disseminate odd-lot information. See MDI Adopting Release, supra note 10, at 18712-13.

the total ongoing, annual burden hours for two exclusive SIPs will be 264 burden hours¹⁸³⁶ and that total ongoing, annual external costs would be \$247,450.¹⁸³⁷

E. Collection of Information is Mandatory

The collection of information discussed above is a mandatory.

F. Confidentiality

This information collection will be public.

G. Revisions to Current MDI Rules Burden Estimates

Currently, the MDI Rules impose “collection of information” requirements within the meaning of the PRA. Specifically, pursuant to Rule 603(b), SROs are required to make available all data necessary to generate consolidated market data to competing consolidators and self-aggregators. As explained in more detail below, the Commission is revising the burden estimates associated with this requirement in light of the amendments. In the MDI Rules, the Commission estimated that each SRO will require an average of 220 initial burden hours of legal, compliance, information technology, and business operations personnel time to prepare and implement a system to collect the information necessary to generate consolidated market data (for a total cost per SRO of \$70,865).¹⁸³⁸ The Commission estimated that each SRO would incur an annual

¹⁸³⁶ The Commission estimates the monetized annual ongoing burden for this requirement to be \$100,602, broken down as follows: [(Sr. Programmer at \$399/hour for 63 hours) + (Sr. Systems Analyst at \$343/hour for 54 hours) + (Compliance Manager at \$373/hour for 6 hours) + (Director of Compliance at \$588/hour for 3 hours) + (Compliance Attorney at \$440/hour for 6 hours) x (2 exclusive SIPs)] = 264 total ongoing, annual burden hours across the exclusive SIPs.

¹⁸³⁷ The Commission estimates total annual ongoing external costs as follows: annual ongoing external costs of \$123,725 per exclusive SIP x (2 exclusive SIPs) = \$247,450.

¹⁸³⁸ In the MDI Adopting Release, the Commission estimated the monetized initial burden for this requirement to be \$70,865. The Commission derived this estimate based on per hour figures from SIFMA’s Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Compliance Manager at \$310 for 105 hours) + (Attorney at \$417 for 70 hours) + (Sr.

average burden on an ongoing basis of 396 hours to collect the information necessary to generate consolidated market data required by Rule 603(b) (for a total cost per SRO of \$128,064).¹⁸³⁹

As described above, the amendments to Rule 603(b) require SROs to make available all data necessary to generate odd-lot information to the exclusive SIPs whereas, under the decentralized consolidation model set forth in the MDI Rules, consolidated market data would be provided by competing consolidators and self-aggregators. The SROs already provide certain quotation information to the exclusive SIPs, and many SROs already provide odd-lot quotation information to customers through their proprietary data feeds.¹⁸⁴⁰ Nevertheless, providing the exclusive SIPs with the data necessary to generate odd-lot information may entail additional burdens. Specifically, technical development work may be needed to direct odd-lot quotations to the exclusive SIPs and to expand the capacity of the existing connections (including acquiring the necessary hardware, network capabilities and power) through which the SROs provide data to the exclusive SIPs to support the additional message traffic associated with odd-lot quotations. Therefore, the Commission is revising its burden estimates for Rule 603(b) upwards by 5% to account for the provision of the data necessary to generate odd-lot information to the exclusive

Systems Analyst at \$285 for 20 hours) + (Operations Specialist at \$137 for 25 hours)] = 220 initial burden hours and \$70,865.

¹⁸³⁹ In the MDI Adopting Release, the Commission estimated the monetized ongoing, annual burden for this requirement to be \$128,064. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Compliance Manager at \$310 for 192 hours) + (Attorney at \$417 for 48 hours) + (Sr. Systems Analyst at \$285 for 96 hours)] = 336 initial burden hours and \$128,064.

¹⁸⁴⁰ See MDI Adopting Release, supra note 10, at 18599.